

"MAXIMA & MINIMA"

Maxima and Minima of func of two variables:-

Defⁿ: A func f of two variables is said to have relative maximum at a point (x_0, y_0) if there is a disc centered at (x_0, y_0) such that $f(x_0, y_0) \geq f(x, y)$ for all point (x, y) that lie inside the disc and f is said to have an absolute maximum at (x_0, y_0) if $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) in the domain of f .

Defⁿ: A func f of two variables is said to have relative minimum at a point (x_0, y_0) if there is a disc centered at (x_0, y_0) such that $f(x_0, y_0) \leq f(x, y)$ for all point (x, y) that lie inside the disc and f is said to have an absolute minimum at (x_0, y_0) if $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) in the domain of f .

Theorem: Let f be a func of two variables with second-order p.d. in some disc centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

- (a) If $D > 0$ and $f_{xx}(x_0, y_0) > 0$ then f has a relative minimum at (x_0, y_0) .
- (b) If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has a relative maximum at (x_0, y_0) .
- (c) If $D < 0$ then f has a saddle point at (x_0, y_0) .
- (d) If $D = 0$ then no conclusion can be drawn.

critical point	f_{xx}	f_{yy}	f_{xy}	D
(0,0)	1	1	0	1
(1,1)	1	1	0	1
(-1,-1)	1	1	0	1

At $(1,1)$ and $(-1,-1)$ we have $D > 0$ and $f_{xx} > 0$ so relative minima occur at these critical points. At $(0,0)$ there is a saddle point since $D > 0$ and $f_{xx} = 0$.

Ex 8 Locate all relative extrema and saddle points of
 $f(x, y) = 4xy - x^4 - y^4$

Soln Since, $f(x, y) = 4xy - x^4 - y^4$
 $f_x(x, y) = 4y - 4x^3$
 $f_y(x, y) = 4x - 4y^3$

So, the critical points of f have co-ordinates satisfying the eqns,

$$4y - 4x^3 = 0, \quad 4x - 4y^3 = 0$$

$$\Rightarrow y = x^3 \quad \text{--- (i)} \quad x = y^3 \quad \text{--- (ii)}$$

Substituting (i) in (ii), we get

$$x = (x^3)^3 = x^9$$

$$\Rightarrow x(x^8 - 1) = 0$$

$$\Rightarrow x = 0, -1, 1$$

$$\text{Hence } y = 0, -1, 1$$

The critical pts are $(0, 0), (-1, -1), (1, 1)$.

Now, $f_{xx}(x, y) = -12x$, $f_{yy}(x, y) = -12y$, $f_{xy}(x, y) = 4$

which yields the following table:

critical points (x, y)	$f_{xx}(x, y)$	$f_{yy}(x, y)$	$f_{xy}(x, y)$	$D = f_{xx}f_{yy} - (f_{xy})^2$
(0, 0)	0	0	4	-16
(-1, -1)	-12	-12	4	128
(1, 1)	-12	-12	4	128

At pt. $(1, 1)$ and $(-1, -1)$ we have $D > 0$ & $f_{xx} < 0$

So relative maxima occur at these critical pts

At $(0, 0)$, there is a saddle point since $D < 0$.

F

Ex:

Locate all relative maxima, relative minima and saddle points, if any

F-19 (a) $f(x, y) = x^2 + y^2 + \frac{2}{xy}$ (b) $f(x, y) = e^x \sin y$

Soln:

Given, $f(x, y) = x^2 + y^2 + \frac{2}{xy}$

$$\Rightarrow f_x(x, y) = 2x - \frac{2}{x^2 y}$$

$$f_y(x, y) = 2y - \frac{2}{xy^2}$$

also, $f_{xx}(x, y) = 2 + \frac{4}{x^3 y}$, $f_{yy}(x, y) = 2 + \frac{4}{xy^3}$

and $f_{xy} = \frac{2}{x^2 y^2}$

The critical pt. occur where $f_x = 0$ and $f_y = 0$

$$2x - \frac{2}{x^2 y} = 0 \quad \text{--- (i)}$$

$$2y - \frac{2}{xy^2} = 0 \quad \text{--- (ii)}$$

(1-1) By subtracting (ii) from (i) we get,

$$2x - \frac{2}{x^2 y} - 2y + \frac{2}{xy^2} = 0$$

$$\Rightarrow 2(x - y) \left(1 + \frac{1}{x^2 y^2} \right) = 0$$

$$\Rightarrow x = y$$

Putting the value of x in (i) we get,

$$2y - \frac{2}{y^3} = 0$$

$$\Rightarrow 2(y^4 - 1) = 0$$

$$\therefore y = 1, y = -1$$

also, $x = 1, x = -1$

$$-\frac{2y}{x^2} + \frac{2x}{xy^2}$$

Hence the critical pt are $(1,1)$ and $(-1,-1)$

We have, $f_{xx} = 2 + \frac{4}{y^3}$, $f_{yy} = 2 + \frac{4}{xy^3}$ and $f_{xy} = \frac{4}{x^2}$

at $(1,1)$.

$$D = (2+4)(2+4) - 2^2 = 36 - 4 = 32 > 0$$

$$\text{also, } f_{xx}(1,1) = 2+4 = 6 > 0$$

So f has a relative minimum at $(1,1)$.

at $(-1,-1)$,

$$D = (2+4)(2+4) - 2^2 = 36 - 4 = 32 > 0$$

$$\text{also, } f_{xx}(-1,-1) = 2+4 = 6 > 0$$

So f has a relative minimum at $(-1,-1)$.

So f has a relative minimum at $(-1,-1)$.

b) Solⁿ

Given $f(x, y) = e^x \sin y$

$$f_x(x, y) = e^x \sin y, \quad f_y = e^x \cos y$$

$$\text{also, } f_{xx} = e^x \sin y, \quad f_{yy} = -e^x \sin y$$

$$\text{and } f_{xy} = e^x \cos y$$

The critical pt. occurs where $f_x = 0$ and $f_y = 0$.

$$\therefore e^x \sin y = 0 \quad \text{and} \quad e^x \cos y = 0$$

$$\sin y = 0, \quad \cos y = 0$$

But $\sin y = \cos y = 0$ is impossible. So there is

no critical points.

Find absolute extrema of the funcⁿ $f(x,y) = xe^y - x^2 - e^y$ on the indicated closed and boundary set R , where R is a rectangular regions with vertices $(0,0)$, $(0,1)$, $(2,1)$, and $(2,0)$.

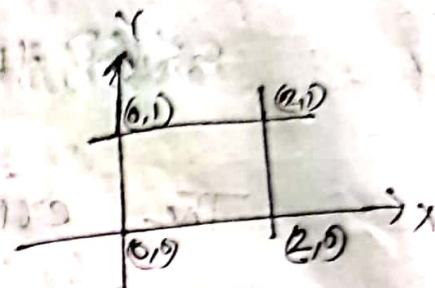
Solⁿ:

The given funcⁿ is,

$$f(x,y) = xe^y - x^2 - e^y \quad \text{--- (i)}$$

$$\Rightarrow f_x = e^y - 2x$$

$$\& f_y = xe^y - e^y = e^y(x-1)$$



Critical pt. occur where $f_x = 0$ and $f_y = 0$.

$$\therefore e^y - 2x = 0 \quad \text{--- (ii)}$$

$$e^y(x-1) = 0 \quad \text{--- (iii)}$$

Using (ii) in (iii) we get,

$$2x(x-1) = 0$$

$$\therefore x = 1$$

$$\text{Putting } x = 1 \text{ in (ii), } e^y = 2$$

$$\therefore y = \ln 2$$

So $(1, \ln 2)$ is the interior critical point.

The boundary consists of four lines.

$$\frac{1}{2} + \frac{1}{4} = 1$$

$$= \frac{2+1}{4} = \frac{3}{4}$$

If f has a relative maximum or a relative minimum at (x_0, y_0) , then we say that f has a relative extremum at (x_0, y_0) and if f has an absolute maximum or absolute minimum at (x_0, y_0) , then we say that f has an absolute extremum at (x_0, y_0) .

Theorem: If $f(x, y)$ is continuous on a closed and bounded set R , then f has both an absolute maximum and absolute minimum on R .

Theorem: If f has relative extremum at a point (x_0, y_0) and if the first P.D. of f exist at this point, then $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$.

Defⁿ: A point (x_0, y_0) in the domain of a func $f(x, y)$ is called a critical point of the func if $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$ or if both P.D. do not exist at (x_0, y_0) .